

AMA3020 - Supplementary

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This document provides some additional or more complete derivations for the theory used in the final report.

Variance of the Monte Carlo estimator and $1/\sqrt{\tau}$ convergence

Recall the estimator

$$\hat{I} = \frac{1}{\tau} \sum_{i=1}^{\tau} h(X_i), \quad h(x) = \frac{f(x)}{p(x)},$$

where $X_i \sim p(x)$ are i.i.d.

Since expectation is linear,

$$\mathbb{E}[\hat{I}] = I.$$

We now compute the variance. Using independence,

$$\text{Var}(\hat{I}) = \text{Var}\left(\frac{1}{\tau} \sum_{i=1}^{\tau} h(X_i)\right) \tag{1}$$

$$= \frac{1}{\tau^2} \sum_{i=1}^{\tau} \text{Var}(h(X_i)) \tag{2}$$

$$= \frac{\tau \text{Var}(h(X))}{\tau^2} \tag{3}$$

$$= \frac{\text{Var}(h(X))}{\tau}. \tag{4}$$

Therefore the standard deviation (standard error) is

$$\sigma_{\hat{I}} = \sqrt{\text{Var}(\hat{I})} = \frac{\sigma_h}{\sqrt{\tau}}.$$

Hence the Monte Carlo error decreases as

$$\sigma_{\hat{I}} \propto \tau^{-1/2}.$$

Variance in importance sampling

The variance of $\hat{I}$ is

$$\text{Var}(\hat{I}) = \frac{1}{\tau} \text{Var}(h(X)).$$

Now

$$\text{Var}(h(X)) = \mathbb{E}[h(X)^2] - \mathbb{E}[h(X)]^2 \tag{5}$$

$$= \int_{\Omega} \left(\frac{f(x)}{p(x)}\right)^2 p(x) dx - I^2 \tag{6}$$

$$= \int_{\Omega} \frac{f(x)^2}{p(x)} dx - I^2. \tag{7}$$

Thus

$$\text{Var}(\hat{I}) = \frac{1}{\tau} \left(\int_{\Omega} \frac{f(x)^2}{p(x)} dx - I^2 \right)$$

This shows the variance depends strongly on the choice of $p(x)$, and is minimized when $p(x)$ is large where $|f(x)|$ is large, motivating importance sampling.

Boltzmann implies Gaussian

In the canonical ensemble,

$$p(q) \propto e^{-\beta U(q)}, \quad \beta = \frac{1}{k_B T}.$$

For the harmonic oscillator,

$$U(q) = \sum_{i=1}^K \frac{1}{2} k q_i^2.$$

Therefore

$$p(q) \propto \exp\left(-\beta \sum_{i=1}^K \frac{1}{2} k q_i^2\right) \quad (8)$$

$$= \prod_{i=1}^K \exp\left(-\frac{\beta k}{2} q_i^2\right). \quad (9)$$

Compare with a normal density

$$\mathcal{N}(0, \sigma^2) : \quad \phi(q) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{q^2}{2\sigma^2}\right).$$

Matching exponents,

$$\frac{1}{2\sigma^2} = \frac{\beta k}{2} \quad \Rightarrow \quad \sigma^2 = \frac{1}{\beta k} = \frac{k_B T}{k}.$$

Hence

$$q_i \sim \mathcal{N}\left(0, \frac{k_B T}{k}\right).$$

Analytical average energy

Since q_i is Gaussian with variance $\sigma^2 = k_B T/k$,

$$\langle q_i^2 \rangle = \sigma^2 = \frac{k_B T}{k}.$$

The potential energy is

$$U(q) = \sum_{i=1}^K \frac{1}{2} k q_i^2.$$

Therefore

$$\langle U \rangle = \sum_{i=1}^K \frac{1}{2} k \langle q_i^2 \rangle \quad (10)$$

$$= \sum_{i=1}^K \frac{1}{2} k \frac{k_B T}{k} \quad (11)$$

$$= \frac{K}{2} k_B T. \quad (12)$$

Metropolis–Hastings acceptance rule

Let $T(x \rightarrow x')$ be the transition probability of the Markov chain:

$$T(x \rightarrow x') = q(x'|x)A(x \rightarrow x'),$$

where q is the proposal distribution and A the acceptance probability.

To sample from $p(x)$ we require detailed balance:

$$p(x)T(x \rightarrow x') = p(x')T(x' \rightarrow x).$$

Substituting,

$$p(x)q(x'|x)A(x \rightarrow x') = p(x')q(x|x')A(x' \rightarrow x).$$

Let

$$r = \frac{p(x')q(x|x')}{p(x)q(x'|x)}.$$

Choose

$$A(x \rightarrow x') = \min(1, r).$$

Then:

If $r \geq 1$:

$$A(x \rightarrow x') = 1, \quad A(x' \rightarrow x) = \frac{1}{r},$$

and detailed balance holds.

If $r < 1$:

$$A(x \rightarrow x') = r, \quad A(x' \rightarrow x) = 1,$$

and detailed balance also holds.

Thus the stationary distribution of the Markov chain is $p(x)$.